# Grey Box: Replicating the Hidden State of Closed Agents from Their Transcripts, to Predict and Prevent Violations

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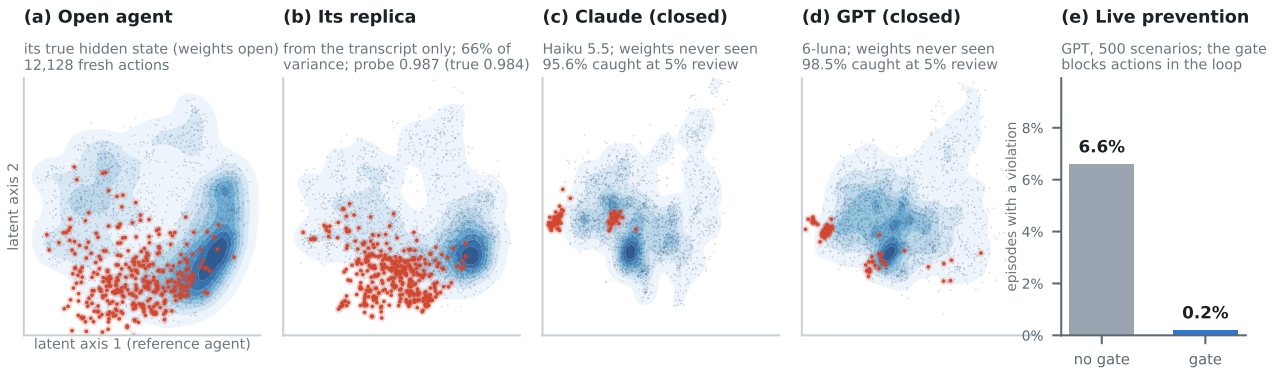



*Figure 1.* **Seeing inside closed agents.** All panels share the two main axes of one reference open agent's hidden state (Gemma-4, at the moment before it acts). Blue: density of safe actions; red: violations of the policy. (a) The open agent's true hidden state on 12,128 fresh actions. (b) The grey-box replica of the same actions, computed from the transcript and the proposed action only (Section 4.2). (c, d) Actions of two closed agents, replicated in the same way. We never saw their weights and fit nothing on them; their own coordinates are not identifiable (Proposition 1), so we show them in the reference agent's coordinates. In every agent the violations occupy their own region, and a monitor on the replica catches 96% and 98% of them when 5% of the actions go to review. (e) Live prevention on GPT-6-luna (T16): the share of episodes with an executed violation, without and with the gate.

## Abstract

Safety tools that read a model's hidden state need its weights, and the agents with the most capability are closed. We ask whether an outside observer can read the hidden state of a closed agent, replicate it, and use the replica to predict and prevent policy violations. The raw coordinates of a hidden state are not identifiable from outputs: a rotation of the hidden space changes no output. Two objects are identifiable, and we build a "grey box" from each. The first is the behavioural state: the model's answers to fixed questions asked in side branches. It recovers the hidden state of models with known internals (Othello-GPT: $R^2$ 0.967), but as a safety signal it only ties a text moderator. The second follows from a simple theorem: an agent's internal state is a function of its tokens, so an open model that reads the same tokens can hold a replica of it. From the transcript alone, a replica in an open reader's latent space reconstructs 66% of the variance of an agent's hidden state, and white-box readouts work on the replica as well as on the true state (AUROC 0.987 vs 0.984). For a policy check on a proposed action, the transcript and the action are Bayes-sufficient. Trained only on an open agent, the replica catches 96% and 98% of the violations of two closed agents, Claude Haiku 5.5 and GPT-6-luna, at a 5% review budget. In the loop of a closed agent, it blocks violations before they run (episodes with an executed violation 6.6% $\rightarrow$ 0.2%). The limit is the coverage of policies, not the agent: labels of a new policy, collected on open agents, transfer to closed ones. Every test is preregistered and hash-locked; we report the failures. We release an SDK and a website on which anyone can train and audit a monitor.

## 1. Introduction

Interpretability can read a model's hidden state: what it knows, and what it will do (Burns et al., 2023; Marks & Tegmark, 2023; Zou et al., 2023; Arditi et al., 2024; Goldowsky-Dill et al., 2025). For agents that act through

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tools, a probe on the hidden state can forecast an unauthorized action before it runs and stop it (Fayyazi et al., 2026). But these tools need the weights, and the agents with the most capability are closed. Their providers give sampled text, and at most top-$k$ log probabilities and a summary of the reasoning.

We ask three questions. **(1)** Can an outside observer read the hidden state of a closed model? **(2)** Can it replicate the state well enough that white-box tools work on the replica? **(3)** Can the replica predict and prevent the agent's violations, in real time?

**Answers.**

1. *What can be read.* The raw coordinates of a hidden state are not identifiable from outputs (Proposition 1). The behavioural state is, and a battery of side-branch questions recovers it on models with known internals (Section 4.1). As a safety signal this "interrogation state" is weak.
2. *What can be replicated.* An agent's internal state is a function of the tokens that it read (Proposition 2). An open reader model on the same tokens replicates 66% of the agent's hidden-state variance, and readouts trained on the true state work on the replica (Section 4.2).
3. *What it is good for.* For a policy check on a proposed action, the transcript and the action are Bayes-sufficient (Corollary 1). A monitor on the replica, trained only on an open agent, catches most violations of two closed agents before they run, and blocks them live (Sections 4.4 and 4.6). Its limit is policy coverage (Section 4.5).

**Rigor.** Every confirmatory test was preregistered and hash-locked before its data existed or was scored. Table 1 lists 18 agent tests with their outcomes, including the ones that failed or could not be tested.

## 2. Theory

**Setting.** A monitor acts at step $t$, after the agent proposes an action and before the action runs. $Z_t$ is every token that the agent read or wrote up to $t$; $V_t$ is the part that a deployment sees, and $H_t$ is the rest (e.g. hidden reasoning). $A_t$ is the proposed action, and $W$ are world facts that the agent did not observe. An internal representation is $\phi_t = \Phi(Z_t; \theta)$, a deterministic function of the tokens.

**Proposition 1** (gauge)**.** *For any invertible $G$, the model whose hidden space is transformed by $G$ (with the adjacent weights compensated) computes the same function. Thus no method that uses only outputs can recover hidden coordinates, and a probe fit in one basis does not transfer to another. The behavioural state (the law of future outputs) is unchanged.*

**Proposition 2** (access is not information)**.** *For every label $Y$, $\sigma(Z_t, \phi_t) = \sigma(Z_t)$. Thus the best monitor that uses $(Z_t, \phi_t)$ is not better than the best monitor that uses $Z_t$, under every strictly proper score and at every point of the ROC curve. Under log loss, the best monitor that uses $Z_t$ beats the best that uses $V_t$ by exactly $I(Y; H_t \mid V_t)$.*

**Corollary 1** (the proposed action is sufficient)**.** *Let $Y = g(A_t, V_t, W)$ with $W \perp\!\!\!\perp Z_t^- \mid V_t$, where $Z_t^-$ are the tokens before the action is sampled, and with sampling noise independent of $(Z_t^-, W)$. Then $P(Y \mid Z_t^-, A_t) = P(Y \mid V_t, A_t)$, and every monitor that reads internals at or before the action has a Bayes risk at least that of the best monitor that reads $(V_t, A_t)$.*

Proofs are in Section A. Proposition 1 says what a grey box can target: a behavioural state, or a state defined up to a learned map. It is the output-side counterpart of identifiability results for representations: latents that supervised training makes identifiable are recovered by encoders that are linear in them when the representation is complete (Zhou & Klindt, 2026), which is why we use a linear map between two models that read the same tokens. Proposition 2 says why a replica is possible: the hidden state is a function of tokens that an outside model can also read. Corollary 1 says when nothing is lost: a probe read before the action must also predict which action the agent samples; a monitor that reads the action does not. It does not cover labels about later behaviour or about beliefs; for later behaviour, the rollout forecast dominates every readout of the state (Section A). What remains is statistical: learning $P(Y \mid V_t, A_t)$, which depends on the labels, the representation, and the coverage of the policies.

## 3. Setup

**Agents and data.** CAPE-Bench (Fayyazi et al., 2026) gives matched agent tasks in which one fact decides if an action is authorized; an environment rule checker labels each action. Open agents: Gemma-4 (26B) and gpt-oss-20b. Closed agents, through their public tool-use APIs, with reasoning off: Claude Haiku 5.5 and GPT-6-luna. External benchmark: AgentDojo prompt injections (Debenedetti et al., 2024).

**The two grey boxes.** *Interrogation state* $\Psi$: the probabilities of the answer "yes" to a fixed battery of questions, each asked in a side branch, so the real conversation never sees it. *Reader replica*: Qwen3-8B reads the agent's instructions, the user's task, the earlier tool calls with their results, and the proposed action. A ridge map from its layer-32 residual stream gives the replica of the agent's state, and a linear probe on the same layer gives the monitor. The reader is trained only on Gemma-4 CAPE-Bench steps (17,623) and is never fit on a test agent.

**Baselines.** The agent's own internals (CAPE's probe); a zero-shot 70B judge (Llama-3.3-70B, same input); TF-IDF with the reader's input and labels; the action's tool alone.

**Statistics.** AUROC within strata (template, tool or suite) with 95% intervals from a bootstrap over scenarios or episodes (1,000 resamples), and the share of violations caught when 5% of actions go to review.

# 4. Results

## 4.1. Reading the state: the interrogation state

On models with known internals, $\Psi$ recovers the state. A battery of 21 questions separates all 21 states of a program compiled into a transformer; after a random rotation of the residual stream, a readout that uses the known layout falls from $r = 0.9995$ to $-0.46$, and $\Psi$ is unchanged (Proposition 1). On Othello-GPT, 61 queries recover the board-carrying hidden state at $R^2$ 0.967 on 1,000 fresh games (preregistered; input-only 0.873). On eight open chat models, the answers recover the hidden state better than a text embedding ($+0.120$ $R^2$, 8/8 models, preregistered), and readouts trained only on open models transfer to Claude (harmful request 0.993, refusal 0.997 AUROC). **As a safety signal, $\Psi$ is weak.** On agents, the answers alone do not match a white-box probe (T1). As a forecast of Claude's refusals, it only ties a graded text moderator in a prospective test (0.873 vs 0.855); with a moderator it doubles the precision of an alarm (Section B).

## 4.2. Replicating the state: the reader replica (T15)

On Gemma-4, where the true state is known, we fit a ridge map from the reader's latent space to the agent's hidden state at the decision point (5 layers, top 128 principal components, 80.5% of the variance) on development steps. On 12,128 fresh steps, the replica reconstructs 66.4% of the variance from the transcript alone; text features reconstruct 58.6% ($+0.079$ [0.075, 0.083]; Figure 2). A probe trained on the agent's *true* states scores 0.984 on true fresh states and 0.987 on the replica ($+0.003$ [$-0.002$, $+0.007$]): white-box readouts work on the replica. The replica is a conditional mean, so it is less spread than the true state, and a text replica also supports the readout (0.983): the part of the state that decides a violation is largely in the tokens, as Proposition 2 implies. A nonlinear map (an MLP) replicates less than the linear one (0.558 vs 0.664, exploratory), as expected when the reader represents the agent's main latents completely (Zhou & Klindt, 2026). Viewing attention as an associative memory over the tokens (Wang, 2026) predicts more interference in longer transcripts. Replication does fall with length (from 0.693 to 0.478 across quintiles of context length), but the probe on the replica stays at the level of the true state in every quintile (exploratory): the part of the state that decides a violation survives.

## 4.3. Why access does not help (T2–T5)

**Timing.** A probe on the agent's internals scores 0.841 at the start of the turn and 0.979 after the reasoning (T2): the white-box advantage over turn-start monitors is when it reads. **Labels.** Resampling scored by a 70B judge gets 0.782, below the probe (T3); the label function is the bottleneck. With labels, a probe on the reader scores 0.994, above the agent's own internals (0.979; $+0.015$ [0.007, 0.026]; T4), and it is non-inferior on unseen task families (0.969 vs 0.976; T5). Given the action, the agent's hidden reasoning adds nothing: $-0.0014$ [$-0.0046$, $+0.0014$] AUROC (exploratory), as Corollary 1 predicts. **Representation.** TF-IDF with the same labels equals the reader on the training benchmark, but is worse on unseen families (0.950 vs 0.969) and at chance on AgentDojo (0.503 vs 0.781; T9).

## 4.4. Two closed agent families (T8b, T11)

Nothing was fit on the closed agents (Table 2). On Claude, the monitor catches 95.6% of 68 violations at a 5% budget; the judge catches none. On GPT-6-luna all three preregistered endpoints pass: pooled 0.996 vs 0.778 ($+0.217$ [0.174, 0.265]); against tool identity alone $+0.141$ [0.111, 0.171]; within tool $+0.039$ [0.015, 0.066]. Seen through Gemma's latent space (the T15 map and the true-state probe), Claude's actions score 0.959 and GPT's 0.976 (exploratory). The probabilities do not transfer between agents (on Claude a 0.5 cut catches 3%), but the ranking does: a threshold at the top 5% of the agent's own unlabelled traffic, then frozen, catches 95.4% online on Claude and 98.4% on GPT, and stays at 94% in a simulated burst of attacks, where a rolling threshold falls to 36%.

## 4.5. The limit is policy coverage (T10, T12–T14)

For a service-restart policy removed from training, the monitor still ranks Claude's violations well (0.912 vs 0.570), but escalates 0 of 26 at a 5% budget; 20 labelled steps of the policy give 85%, 40 give 94% (T10, exploratory curve; Figure 4a). A held-out credentials policy with similar trained policies transfers with no labels (T12: the preregistered H1 fails, because there was nothing to fix). Prompt injection on AgentDojo is a new policy: on GPT-6-luna, the monitor is not significantly better than the judge (0.840 vs 0.787; T13b fails). We added labelled AgentDojo calls of the two *open* agents and tested on the *closed* agent (T14): 0.922 vs 0.787 ($+0.136$ [0.078, 0.191]); 92% of the injected calls caught, against 36% without these labels (Figure 4b). Labels of a policy transfer from open agents to closed ones.

## 4.6. Live prevention on a closed agent (T16)

We ran the monitor as a server in the loop of GPT-6-luna on 500 CAPE-Bench scenarios, each with and without the

*Table 1.* All agent tests, in the order in which they were locked. Pass: the preregistered criterion was met.

| Test | Question | Key result | Outcome |
|---|---|---|---|
| T1 | Resampling vs a probe at the turn start (oracle checker) | 0.904 vs 0.841 | pass; answers alone: fail |
| T2 | Internals after the reasoning vs monitors at the turn start | 0.979 vs 0.904 | pass |
| T3 | Resampling with a deployable 70B judge | 0.782 $<$ probe 0.846 | fail (probe better) |
| T4 | Reader with labels vs the agent's internals | 0.994 vs 0.979 | pass |
| T5 | The same, on unseen task families | 0.969 vs 0.976 (non-inferior) | pass |
| T6 | Reader trained on Gemma, applied to gpt-oss | 0.915 vs judge 0.833 | pass |
| T7 | Reader on AgentDojo, no AgentDojo training | 0.781 vs judge 0.537 | pass |
| T8 | Claude, actions after the reveal | 0 violations in the units | not tested |
| T8b | Claude, all actions | 95.6% caught vs 0% | pass |
| T9 | TF-IDF with the same labels vs the reader | equal in distribution; chance on AgentDojo | mixed |
| T10 | Claude, a policy family removed from training | 0.912 vs 0.570; 0/26 caught | pass (ranking) |
| T11 | GPT-6-luna, beyond tool identity | 0.996 vs 0.778; tool 0.856 | pass |
| T12 | Labels for a new policy, prospective | 18/18 caught with 0 labels | H1 fail |
| T13 | GPT-6-luna on AgentDojo, with spotlighting | attack success 2.1%; 4 injected calls | not tested |
| T13b | The same, without the defense | 0.840 vs judge 0.787 | fail |
| T14 | T13b after adding AgentDojo labels of open agents | 0.922 vs judge 0.787; 92% caught | pass |
| T15 | Does the reader replicate the agent's hidden state? | $R^2$ 0.664 (text 0.586); readout 0.987 vs 0.984 | pass |
| T16 | Live prevention: the gate blocks GPT-6-luna's actions | 6.6% $\rightarrow$ 0.2% of episodes | pass |

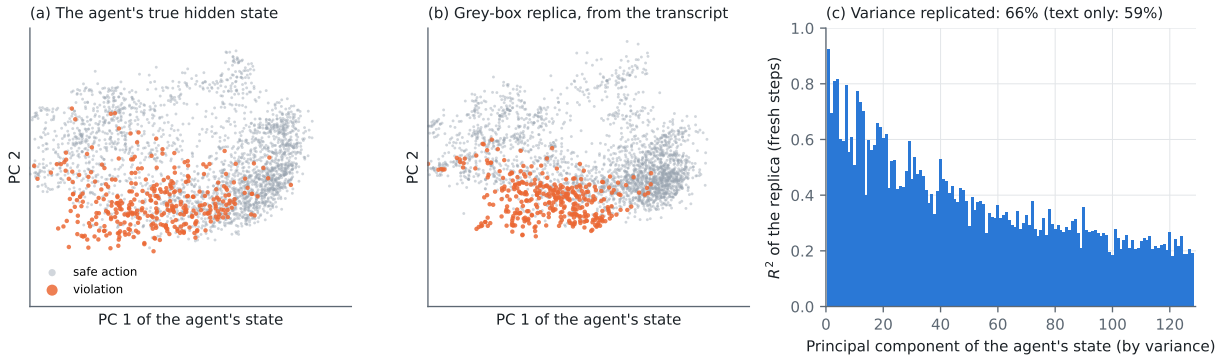



*Figure 2.* Replicating the hidden state of an agent from its transcript (T15, 12,128 fresh steps). (a) The agent's true decision-point state, on its first two principal components. (b) The grey-box replica of the same steps, from the transcript and the proposed action only, on the same axes. (c) The share of variance that the replica reconstructs, per principal component.

*Table 2.* Closed agents; monitors trained only on Gemma-4. Caught: violations escalated when 5% of the actions go to review.

| | Claude Haiku 4.5 | | GPT-6-luna | |
|---|---|---|---|---|
| | AUROC | caught | AUROC | caught |
| Reader | 0.989 | 95.6% | 0.996 | 98.5% |
| TF-IDF | 0.997$^\dagger$ | 64.7% | 0.964 | 50.0% |
| Zero-shot judge | 0.570 | 0% | 0.778 | 37.9% |
| Tool identity | – | – | 0.856 | 60.6% |

Pooled AUROC. $^\dagger$within the one stratum with both classes.

gate (preregistered). The threshold came from GPT's past, unlabelled traffic. A blocked action does not run; the agent is told that it went to a human. The share of episodes with an executed violation fell from 6.6% to 0.2% (-0.064 [-0.088, -0.044]; P1 passes). The gate blocked 67 actions, of which 36% would have been violations; in safe scenarios it blocked 0.08 actions per episode. The agent executed 0 violations after a block. Task goals were reached in 47.4% (no gate) and 45.6% (gate) of the episodes; in these scenarios a goal can require the unauthorized action. The server round trip was 89 ms (median).

## 5. The tool

We release an SDK, a monitoring server and a website (links in the camera-ready version). On the website, a user imports their agent's logs (OpenAI or Anthropic tool-calling format), labels actions, trains a monitor in the browser, and gets a cross-validated accuracy over held-out conversations, a frozen review threshold and the catch rate at their budget; the logs stay in the browser. A review queue ranks the unlabelled actions by risk: on 300 GPT-6-luna conversations with half of them labelled, its first 25 actions were all violations (base rate 4.7%), and its first 43 held 92% of

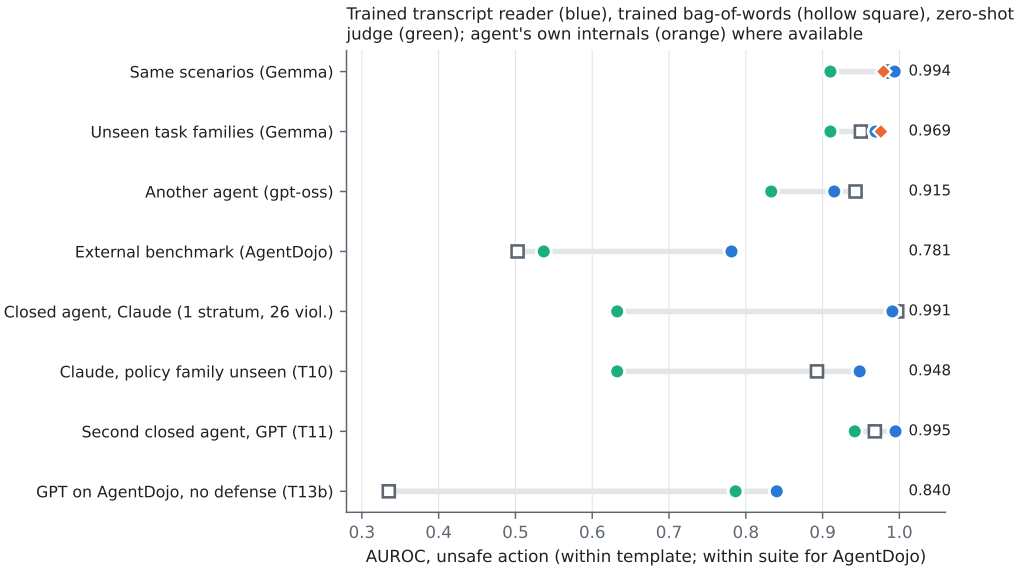



*Figure 3.* Monitoring the next action from the transcript, under increasing shift: the reader (blue), TF-IDF with the same labels (hollow square), a zero-shot 70B judge (green), and the agent's own internals where available (orange). The two Claude rows rest on one stratum (136 actions, 26 violations). In the T10 row the monitors were retrained without that policy; in the T13b row the policy (prompt injection) is new.

the violations. A guard places the monitor in an OpenAI or Anthropic tool loop. The browser and Python give byte-identical text on 302 real agent steps and the same held-out AUROC (0.9887 vs 0.9886). The text monitor takes 0.18 ms per action; the reader takes 41 ms on one GPU, and the server round trip in the loop of a closed agent was 89 ms (median; 95th percentile 215 ms). A live viewer shows a closed chat model's interrogation state in a 3D space shared with open models, at about $0.001 per turn.

## 6. Related work

**Probes on internals** read truth, refusal and deception (Burns et al., 2023; Marks & Tegmark, 2023; Zou et al., 2023; Arditi et al., 2024; Goldowsky-Dill et al., 2025); CAPE probes an agent before each action (Fayyazi et al., 2026). **Chain-of-thought monitors** catch reward hacking that action-only monitors miss (Baker et al., 2025; Korbak et al., 2025; Chen et al., 2025); Corollary 1 says that for labels that the action and the visible facts decide, the reasoning adds nothing once the monitor sees the action. **AI control** audits a fixed budget of actions with a trusted monitor (Greenblatt et al., 2024); we show that the threshold must come from the monitored agent's own traffic. **Identifiability.** Hidden coordinates are identifiable only up to transformations that leave the function unchanged; supervised learning identifies latents linearly when the representation is complete and through sparsity otherwise (Zhou & Klindt, 2026). Our gauge result applies the same logic to an observer who sees only outputs. **Black-box access** recovers parts of a model from logits (Carlini et al., 2024; Finlayson et al., 2024), detects lies with follow-up questions (Pacchiardi et al., 2024; Cooney et al., 2026; Sam et al., 2025), forecasts behaviour from traces (Levy et al., 2026), and probes closed transcripts with open readers (conti et al., 2026). We add the identifiability and sufficiency theory, a direct test that the reader replicates the hidden state, closed agents from two providers, live prevention, and a preregistered test of each claim.

## 7. Limitations

- We can test replication only where the true state is known (open agents). For closed agents, we test what the replica is for: prediction and prevention.
- Most agent tests use one benchmark. On AgentDojo, a monitor without labels of the new policy is not significantly better than a judge on a closed agent (T13b).
- Policy coverage was measured on three held-out policies; the role of similarity is a hypothesis.
- Corollary 1 needs a label that the action and the visible facts decide. It does not cover beliefs or later behaviour.
- The interrogation state is weak as a safety signal; the reader replica carries the safety results.

## 8. Conclusion

The hidden state of a closed agent cannot be read in its own coordinates, but it can be replicated: an open model that reads the same tokens holds most of it, and white-box readouts work on the replica. For the actions of closed

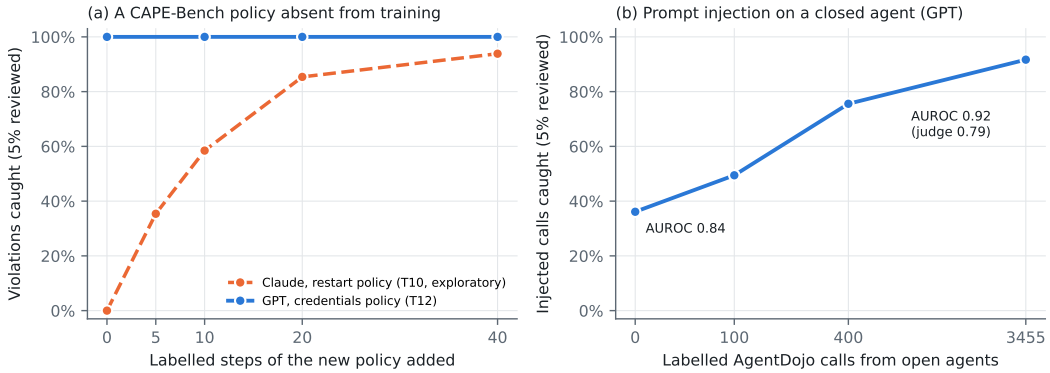



*Figure 4.* Policy coverage. (a) A CAPE-Bench policy removed from training, against the number of its labelled steps added back (T10 exploratory; T12 preregistered). (b) Prompt injection on the closed GPT agent, against labelled AgentDojo calls from *open* agents (T14, preregistered).

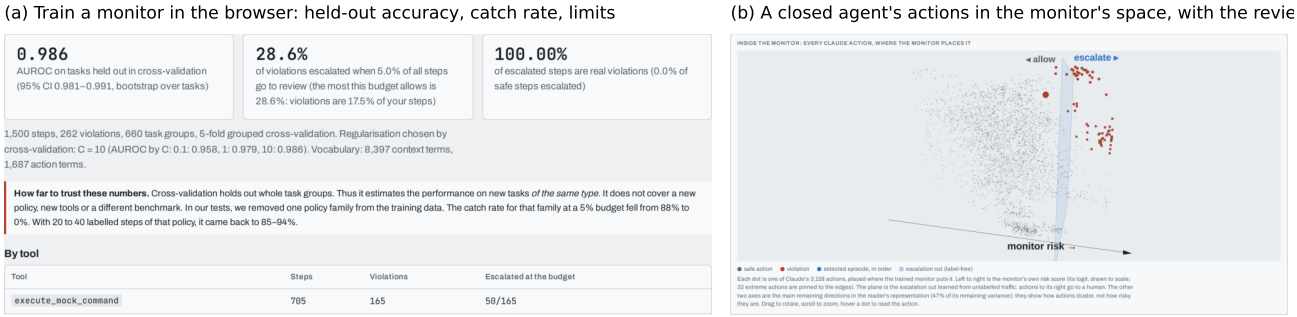



*Figure 5.* The released tool. (a) The browser trainer on 1,500 example steps: cross-validated AUROC over held-out tasks with its interval, the catch rate at the user's review budget with its ceiling, and the limits of the estimate. (b) The replay of Claude Haiku 5.5 as an agent: each of its 3,128 actions in the monitor's space, with the label-free review cut as a plane.

agents, this replica predicts and prevents violations; the limit is labels of each policy, which transfer from open agents to closed ones.

## References

Arditi, A., Obeso, O., Syed, A., Paleka, D., Panickssery, N., Gurnee, W., and Nanda, N. Refusal in language models is mediated by a single direction, 2024.

Baker, B., Huizinga, J., Gao, L., Dou, Z., Guan, M. Y., Madry, A., Zaremba, W., Pachocki, J., and Farhi, D. Monitoring reasoning models for misbehavior and the risks of promoting obfuscation. *arXiv preprint arXiv:2503.11926*, 2025.

Burns, C., Ye, H., Klein, D., and Steinhardt, J. Discovering latent knowledge in language models without supervision. In *International Conference on Learning Representations (ICLR)*, 2023.

Carlini, N., Paleka, D., Dvijotham, K. D., Steinke, T., Hayase, J., Cooper, A. F., Lee, K., Jagielski, M., Nasr, M., Conmy, A., Yona, I., Wallace, E., Rolnick, D., and Tramèr, F. Stealing part of a production language model, 2024.

Chen, Y., Benton, J., Radhakrishnan, A., Uesato, J., Denison, C., Schulman, J., Somani, A., Hase, P., Wagner, M., Roger, F., Mikulik, V., Bowman, S. R., Leike, J., Kaplan, J., and Perez, E. Reasoning models don't always say what they think. *arXiv preprint arXiv:2505.05410*, 2025.

conti, ShivamR, Sharf, J., jain18ayush, and johnyan. Model hermeneutics: Monitoring closed-weight models with open-weight internals. LessWrong, 30 September 2026, 2026. URL https://www.lesswrong.com/posts/CZLhfNGaDqtYxvGJD/model-hermeneutics-monitoring-closed-weight-model

Cooney, A., Africa, D., and Irving, G. "did you lie?" evalu-

ating lie detectors across scale and belief-verified model organisms, 2026.

Debenedetti, E., Zhang, J., Balunović, M., Beurer-Kellner, L., Fischer, M., and Tramèr, F. AgentDojo: A dynamic environment to evaluate prompt injection attacks and defenses for LLM agents. In *Advances in Neural Information Processing Systems (Datasets and Benchmarks Track)*, 2024.

Fayyazi, A., Kamal, M., and Pedram, M. The agent knew: Internal states forecast unauthorized actions and stop them, and the lies that follow, 2026.

Finlayson, M., Ren, X., and Swayamdipta, S. Logits of API-protected LLMs leak proprietary information, 2024.

Goldowsky-Dill, N., Chughtai, B., Heimersheim, S., and Hobbhahn, M. Detecting strategic deception with linear probes. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, volume 267 of *PMLR*, pp. 19755–19786, 2025.

Greenblatt, R., Shlegeris, B., Sachan, K., and Roger, F. AI control: Improving safety despite intentional subversion. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, 2024.

Korbak, T., Balesni, M., Barnes, E., Bengio, Y., Benton, J., Bloom, J., Chen, M., Cooney, A., Dafoe, A., Dragan, A., Emmons, S., Evans, O., Farhi, D., Greenblatt, R., Hendrycks, D., Hobbhahn, M., Hubinger, E., Irving, G., Jenner, E., Kokotajlo, D., Krakovna, V., Legg, S., Lindner, D., Luan, D., Madry, A., Michael, J., Nanda, N., Orr, D., Pachocki, J., Perez, E., Phuong, M., Roger, F., Saxe, J., Shlegeris, B., Soto, M., Steinberger, E., Wang, J., Zaremba, W., Baker, B., Shah, R., and Mikulik, V. Chain of thought monitorability: A new and fragile opportunity for AI safety. *arXiv preprint arXiv:2507.11473*, 2025.

Levy, M., Goldberg, Y., and Cooper Stickland, A. Forecasting future behavior as a learning task, 2026.

Marks, S. and Tegmark, M. The geometry of truth: Emergent linear structure in large language model representations of true/false datasets, 2023.

Pacchiardi, L., Chan, A. J., Mindermann, S., Moscovitz, I., Pan, A. Y., Gal, Y., Evans, O., and Brauner, J. How to catch an AI liar: Lie detection in black-box LLMs by asking unrelated questions. In *International Conference on Learning Representations (ICLR)*, 2024.

Sam, D., Finzi, M., and Kolter, J. Z. Predicting the performance of black-box LLMs through follow-up queries, 2025.

Wang, A. You could have invented transformers from memory. Blog post, 15 October 2026, 2026. URL https://alexwang.ai/posts/associative-memory/.

Zhou, J. and Klindt, D. Complete or sparse: A tale of two identifiabilities, 2026.

Zou, A., Phan, L., Chen, S., Campbell, J., Guo, P., Ren, R., Pan, A., Yin, X., Mazeika, M., Dombrowski, A.-K., Goel, S., Li, N., Byun, M. J., Wang, Z., Mallen, A., Basart, S., Koyejo, S., Song, D., Fredrikson, M., Kolter, J. Z., and Hendrycks, D. Representation engineering: A top-down approach to AI transparency. *arXiv preprint arXiv:2310.01405*, 2023.

## A. Proofs

**Setting.** $Z_t$ are all tokens that the agent read or wrote up to step $t$; $V_t$ is the visible part and $H_t$ the hidden part; $A_t$ is the proposed action; $W$ are world facts that the agent did not observe. An internal representation is $\phi = \Phi(Z; \theta)$, a deterministic function of the tokens (numerical noise in a batch is independent of everything else).

**Proposition 3 (behavioural sufficiency).** Let $Y = g(\tau)$ be defined by the future $\tau$ that the agent and the environment generate from context $x$, and let $\phi(x)$ carry no information about the future sampling noise given $x$. Then $\pi^*(x) = P(Y = 1 \mid x)$ has a maximal ROC curve and a minimal risk under every strictly proper score among all forecasts $f(\phi(x))$.

*Proof.* By the assumption and the tower property, $E[Y \mid \phi(X)] = E[\pi^*(X) \mid \phi(X)]$. Under a strictly proper score, the Bayes act given $\sigma(X)$ is $\pi^*(X)$, and $\sigma(\phi(X)) \subseteq \sigma(X)$; less information cannot raise the expected score. For the ROC curve, the Neyman–Pearson lemma gives the most powerful test of each size as a threshold on the likelihood ratio, which is a strictly increasing function of $\pi^*$; every test based on $f(\phi(X))$ is a test based on $X$. □

**Proposition 1 in the main text (access is not information).** (i) $\phi_t$ is $\sigma(Z_t)$-measurable, so $\sigma(Z_t, \phi_t) = \sigma(Z_t)$, and Proposition 3 with $X = Z_t$ gives the first claim. (ii) The Bayes log loss of predicting $Y$ from a $\sigma$-field $\mathcal{G}$ is the conditional entropy $H(Y \mid \mathcal{G})$; thus $H(Y \mid V_t) - H(Y \mid V_t, H_t) = I(Y; H_t \mid V_t)$. □

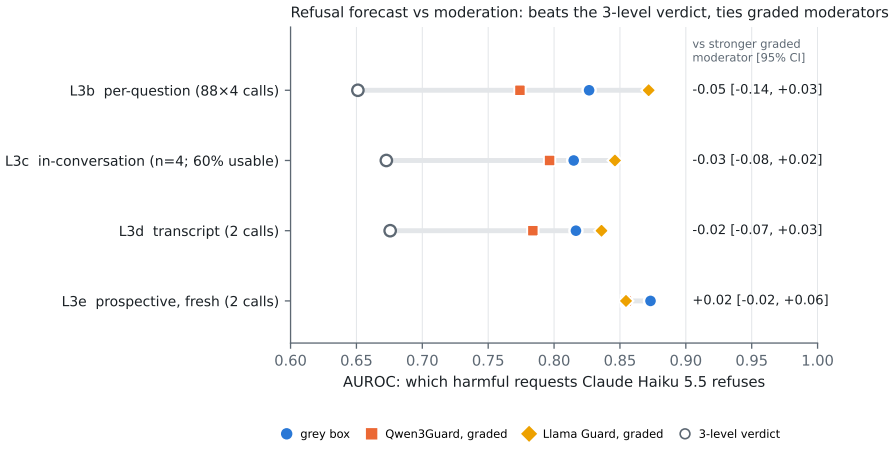
**Corollary 1 in the main text (the proposed action is sufficient).** Let $Z^-$ be the tokens before the action is sampled, $A_t = a(Z^-, \xi)$ with sampling noise $\xi \perp\!\!\!\perp (Z^-, W)$, $W \perp\!\!\!\perp Z^- \mid V_t$, and $Y = g(A_t, V_t, W)$.

1. From $W \perp\!\!\!\perp Z^- \mid V_t$ and $\xi \perp\!\!\!\perp (Z^-, W)$: $W \perp\!\!\!\perp (Z^-, \xi) \mid V_t$.
2. $A_t$ is a function of $(Z^-, \xi)$, so $W \perp\!\!\!\perp (Z^-, A_t) \mid V_t$.
3. By weak union, $W \perp\!\!\!\perp Z^- \mid (V_t, A_t)$.
4. $Y$ is a function of $(A_t, V_t, W)$, so $Y \perp\!\!\!\perp Z^- \mid (V_t, A_t)$, that is, $P(Y \mid Z^-, A_t) = P(Y \mid V_t, A_t)$.
5. A monitor that reads internals at or before the action uses $\phi = \Phi(Z^-)$, with $\sigma(\phi) \subseteq \sigma(Z^-) \subseteq \sigma(Z^-, A_t)$. By step 4 the Bayes predictor given $\sigma(Z^-, A_t)$ equals the one given $(V_t, A_t)$, and less information cannot help (Proposition 3). □

The inequality can be strict: if $A_t$ is random given $Z^-$ and $Y$ depends on which action is sampled, a probe that reads before the action must average over the actions. The assumption on $W$ fails only if the internals hold information about the world that is not in the tokens, which cannot happen for a model whose only input is its token history.

## B. The interrogation state in chat

On Claude Haiku 5.5, readouts trained only on open models forecast which harmful requests Claude refuses. Against a 3-level moderation verdict the forecast wins (preregistered L3b, L3c, L3d); against graded moderators it ties (L3e, prospective, 923 conversations: 0.873 vs 0.855). Combined with a moderator, it doubles the precision of an alarm for harmful requests that the model answers (0.535 vs 0.274).



## C. The full preregistration record

Every confirmatory test, in the order in which it was locked. Locks (SHA-256 of the preregistration and its code) and data are in the released record.

| Test | Outcome (verbatim from the record) |
|---|---|
| L2 open LLMs (H1–H3) | **PASS all, after deviation 1**: H1 +0.120 [0.115, 0.126]; H2 +0.057 [0.045, 0.073] (7/8 models); H3 +0.359 [0.310, 0.415]; Holm p ¡ 0.0005 each. First run: H1 valid PASS, H2/H3 invalid (numerical; see note) (results/l2/l2_confirm.json) |
| T1 agents (F1–F3; addendum 1: F4–F5) | **Claim (a) CONFIRMED** (F1, F2 pass: R16 − WB +0.063 [0.033, 0.094]); claim (b) NOT confirmed (F3 fails: Ψ − WB −0.021 [−0.054, 0.009]); addendum **F4 passes** (R4+Ψ − WB +0.049 [0.020, 0.080]); F5 fails (R2+Ψ − R16 −0.023 [−0.053, 0.006]) (results/t1/t1_agents_confirm.json) |
| L1 Othello recovery (E1–E3) | **PASS all**: E1 $R^2$ 0.967 [0.967, 0.968] ¿ 0.90; E2 $\Delta R^2$ +0.094 [0.086, 0.105], Holm p ¡ 0.0005; E3 Δ occupied acc +0.043 [0.040, 0.046], Holm p ¡ 0.0005. White-box reference 0.983 (results/othello/l1_confirm.json) |
| L3 Claude Haiku 5.5 readout transfer (C1–C3) | **all pass**: C1 0.993 [0.983, 0.999]; C2 0.997 [0.990, 1.000]; C3 0.935 [0.868, 0.982] (results/l3/) |
| L3b Claude Haiku 5.5: grey box vs text moderation (D1–D3) | **D1 pass** 0.923 [0.884, 0.957]; **D2 (key) pass** grey box − moderation +0.176 [0.060, 0.287] (0.827 vs 0.651); D3 not tested (1/40 over-refusals) (results/l3/claude_livebench.json) |
| L3c Claude Haiku 5.5: one-call grey box + compliance alarm (E1–E3) | **all pass** vs the preregistered 3-level verdict: E1 +0.142 [0.081, 0.203] (0.815 vs 0.673); E2 AP +0.230 [0.161, 0.307]; E3 AP +0.081 [0.036, 0.140]. Against GRADED moderators E1 ties (Qwen3Guard +0.018 [−0.043, +0.081]; Llama-Guard −0.031 [−0.082, +0.020]); the alarm gain holds (results/l3/claude_batch.json, l3c_graded_reanalysis.json) |
| L3d Claude Haiku 5.5: transcript-framing grey box (F1–F2) | **both pass**: F1 grey box − moderation on harmful refusals +0.141 [0.082, 0.199] (0.817 vs 0.676); F2 compliance-risk AP (Claude-moderator) +0.115 [0.076, 0.179] (results/l3/claude_transcript.json) |
| T2 time: internals at CAPE's decision point vs black box (G1–G3) | **all pass**: G1 I_dec − R16_ts +0.075 [0.044, 0.106]; G2 R16_dec − I_dec +0.021 [0.011, 0.035]; G3 I_dec − WB_ts +0.138 [0.107, 0.163] (results/t1/t2_time.json) |
| L3e Claude Haiku 5.5: the Live tool's alarm, prospective (A1–A3) | **A1 pass** (alarm precision 0.535 vs 0.274 for the moderator alone, +0.261 [0.215, 0.311]); **A2 FAIL** (grey box 0.873 vs graded Qwen3Guard 0.855, +0.018 [−0.023, 0.062]); **A3 FAIL** (multi-turn 0.894 vs 0.906) (results/l3/l3e_tool.json) |
| T3 deployable monitors: LLM judge instead of the oracle checker (K1–K3) | **K1 FAIL** (judge-scored resampling 0.782 ¡ turn-start probe 0.846, −0.065 [−0.115, −0.019]); **K2: probe better** (CAPE probe 0.976 ¿ judge on actual action 0.910, −0.067 [−0.104, −0.036] for the judge); **K3 FAIL** (resampling hurts the action judge, −0.049) (results/t1/t3_deployable.json) |
| T4 reader trained on labels vs the agent's internals (M1–M2) | **both pass**: M1 READER_act − I_dec +0.015 [0.007, 0.026] (0.994 vs 0.979); M2 READER_act − J_act +0.085 [0.052, 0.127] (results/t1/t4_reader.json) |
| T5 agent internals vs transcript reader on unseen scenario families (P1–P2) | **P1 pass** (reader − agent internals on unseen families −0.007 [−0.014, −0.0001], margin −0.02); **P2: agent probe better than zero-shot judge** (+0.062 [0.028, 0.101]) (results/t1/t5_family.json) |
| T6 reader trained on Gemma, monitoring gpt-oss (Q1) | **Q1 pass**: Gemma-trained reader on gpt-oss 0.915 vs zero-shot judge 0.833, +0.082 [0.043, 0.129] (results/t1/t6_cross_agent.json) |
| T7 reader monitor on AgentDojo prompt injection, zero transfer training (R1–R2) | **R1 pass**: reader 0.781 vs zero-shot judge 0.537, +0.249 [0.136, 0.369]; **R2 reader better than CAPE's gate**: gpt-oss calls as the prereg text specifies, 0.823 vs 0.601, +0.222 [0.143, 0.303] (locked code's broader subset: +0.264 [0.174, 0.359]) (results/t1/t7_agentdojo.json) |
| T8 Claude Haiku 5.5 as an agent, monitored by a reader trained only on Gemma (S1) | **S1 NOT TESTED**: 0 unsafe units under the preregistered post-reveal rule (Claude's 68 violations all happen at or without the reveal); see T8b |
| T8b Claude as an agent, all non-read actions (S1b) | **S1b pass**: Gemma-trained reader on Claude 0.991 vs zero-shot judge 0.633, +0.360 [0.302, 0.413]; 95.6% of violations caught at 5% escalation (judge 0%) (results/t1/t8b_claude_all_actions.json) |
| T9 trained TF-IDF monitor on the reader's own input, same labels (B1 Claude, B2 unseen families) | **B2 pass, B1 FAIL**: unseen families reader 0.969 vs TF-IDF 0.950 (+0.019 [0.007, 0.032]); Claude reader 0.991 vs TF-IDF 0.997 (−0.006 [−0.027, +0.007]). AgentDojo (reported) 0.781 vs 0.503 (results/t1/t9_text_baseline.json) |
| T10 Claude on task families the monitor never saw (C1 pooled, C2 within) | **C1, C2 pass** (vs zero-shot judge): pooled 0.912 vs 0.570 (+0.342 [0.311, 0.371]); within 0.948 vs 0.633 (+0.318 [0.265, 0.374]). But at the label-free cut the unseen restart family is caught 0/26 (results/t1/t10_claude_unseen_families.json) |

| Test | Outcome (verbatim from the record) |
|---|---|
| T11 second closed family: gpt-6-luna as agent, frozen monitors (G1 pooled vs judge, G2 vs tool identity, G3 within tool) | **G1, G2, G3 all pass**: GPT-6-luna, 3,627 actions, 66 unsafe in 6 families. Reader pooled 0.996 vs judge 0.778 (+0.217 [0.174, 0.265]); vs tool identity +0.141 [0.111, 0.171]; within tool vs judge +0.039 [0.015, 0.066]. 98.5% caught at the label-free 5% cut (judge 37.9%) (results/t1/t11_gpt_agent.json) |
| T12 labels needed for a new policy, prospective on GPT (H1 40 vs 0 labels, H2 catch at $40 \geq 0.5$) | **H1 FAILS, H2 passes**: with 0 labels of the held-out credentials policy, the reader already caught 18/18 of GPT's violations of it, so added labels had nothing to fix ($40 - 0$: 0.0). The T10 curve does not replicate on this family (results/t1/t12_new_policy_labels.json) |
| T13 a closed agent on an external benchmark: GPT-6-luna on AgentDojo, frozen monitors (K1 reader vs judge) | **K1 NOT TESTED**: GPT-6-luna resisted the injections (attack success 2.1%), leaving 4 injected calls (minimum 10). Utility on benign tasks 0.862 (results/t1/t13_gpt_agentdojo.json) |
| T13b closed agent on AgentDojo without the defense (arm none, dev + conf; K1b reader vs judge) | **K1b FAILS**: 36 injected calls; reader 0.840 vs zero-shot judge 0.787 within suite (+0.052 [−0.026, +0.130]). TF-IDF 0.335. At the benign-call 5% cut: reader 36%, judge 53% (results/t1/t13b_gpt_agentdojo_none.json) |
| T14 labels of the new policy from open agents, tested on the closed agent (L1 vs judge, L2 vs no labels) | **L1, L2 pass**: with AgentDojo labels from open agents, the reader on GPT's AgentDojo calls scores 0.922 vs judge 0.787 (+0.136 [0.078, 0.191]) and vs no labels +0.081 [0.044, 0.121]; caught at the benign 5% cut 36% → 92% (results/t1/t14_agentdojo_labels.json) |
| T15 latent replication: does the reader replicate the agent's hidden state? (E1 $R^2$ vs text, E2 readouts on the replica) | **E1, E2 pass**: from the transcript alone, the reader replica reconstructs 66.4% of the agent's decision-state variance (top 128 PCs; text 58.6%; +0.079 [0.075, 0.083]); a probe trained on TRUE states scores 0.987 on the replica vs 0.984 on the true state (+0.003 [−0.002, +0.007]) (results/t1/t15_latent_replication.json) |
| T16 live prevention on a closed agent: the monitor blocks GPT-6-luna's escalated actions in the loop (P1 violation episodes, gated − ungated) | pending |